

By Theorem 3 from Chapter 1, the matrix equation $Ax=0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$.

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 0 \\ -6 & -4 & 7 & 8 & 0 \\ -7 & 7 & 5 & 9 & 0 \end{bmatrix} \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ + 6 \times \text{Row 1}}]{\substack{\text{Row 3} = \text{Row 3} \\ + 7 \times \text{Row 1}}} \begin{bmatrix} 1 & 3 & -2 & 0 & 0 \\ 0 & 14 & -5 & 8 & 0 \\ -7 & 7 & 5 & 9 & 0 \end{bmatrix} \xrightarrow[\substack{-2 \times \text{Row 2}}]{\substack{\text{Row 3} = \text{Row 3}}} \begin{bmatrix} 1 & 3 & -2 & 0 & 0 \\ 0 & 14 & -5 & 8 & 0 \\ 0 & 0 & 1 & -7 & 0 \end{bmatrix}$$

It's easy to see that the final matrix is in echelon form.

~~Row~~ 4 has no pivots, i.e. x_4 is a free variable.

By Theorem 2 from Chapter 1, the matrix eqn. $Ax=0$ has infinite solutions. $\therefore$ The columns of A are not linearly independent and hence the vectors do not form a basis of $\mathbb{R}^3$. $\square$

In Exercises 21 and 22, mark each statement True or False. Justify each answer.

- 21) a) A subspace of $\mathbb{R}^n$ is any set H such that (i) the zero vector is in H
 (ii) u, v and $u+v$ are in H , and (iii) c is a scalar and cu is in H .

Solution: The given statement is True by the definition of subspace.
 b) If $v_1, \dots, v_p$ are in $\mathbb{R}^n$, then $\text{Span}\{v_1, \dots, v_p\}$ is the same as the column space of the matrix $\{v_1, \dots, v_p\}$.

Solution: By Defn, the Column space of the matrix $\{v_1, \dots, v_p\}$ is the set of all linear combinations of the columns of the matrix, i.e. $v_1, v_2, \dots, v_p$. This is equivalent to the definition of the $\text{Span}\{v_1, \dots, v_p\}$. $\therefore$ The given statement is True. $\square$

- c) The set of all solutions of a system of m homogeneous equations in n unknowns is a subspace of $\mathbb{R}^n$. Let the set of solutions be S .

Solution: A system of m homogeneous equations in n unknowns is equivalent to solving the matrix equation $Ax=0$ where A is an $m \times n$ matrix. The rows of A represents the coefficients and the columns are the variables.